

Method	Endpoint MSE@5	Endpoint MSE@10	Gain@5 (%)	Gain@10 (%)
Framewise	0.15866	0.21429	reference	reference
Constant dynamics	0.15863	0.21423	+0.02	+0.03
ShiftWM (ours)	0.15830	0.21342	+0.23	+0.40
Action-free	0.15936	0.21496	-0.44	-0.31

Table 1: **Matched h10-trained methods: endpoint mse.** Every method completed 30 epochs with width 192, depth 4 and three support frames, at seeds 0/1/2. Values are means across seeds and equally weighted validation episodes; all four methods use identical populations within each horizon. The standard h5 and h10 populations may differ because each requires a complete query window. Gains use Framewise as the reference within the same horizon. **Bold green** marks positive point reductions, not significance. All methods and regressions are retained; these are development results, not a new test evaluation.

Method	Mean query MSE@5	Mean query MSE@10	Gain@5 (%)	Gain@10 (%)
Framewise	0.11286	0.15261	reference	reference
Constant dynamics	0.11284	0.15258	+0.01	+0.02
ShiftWM (ours)	0.11271	0.15222	+0.13	+0.25
Action-free	0.11323	0.15322	-0.33	-0.40

Table 2: **Matched h10-trained methods: mean query mse.** Every method completed 30 epochs with width 192, depth 4 and three support frames, at seeds 0/1/2. Values are means across seeds and equally weighted validation episodes; all four methods use identical populations within each horizon. The standard h5 and h10 populations may differ because each requires a complete query window. Gains use Framewise as the reference within the same horizon. **Bold green** marks positive point reductions, not significance. All methods and regressions are retained; these are development results, not a new test evaluation.

Scoring	Ours	Framewise	Gain (%)	Paired 95% interval
Endpoint 5	0.15830	0.15866	+0.23	[-0.00052, -0.00022]
Mean 1–5	0.11271	0.11286	+0.13	[-0.00022, -0.00008]
Endpoint 10	0.21342	0.21429	+0.40	[-0.00133, -0.00049]
Mean 1–10	0.15222	0.15261	+0.25	[-0.00059, -0.00022]

Table 3: **ShiftWM (ours) versus equally h10-trained Framewise.** Intervals are for ours minus Framewise in standardized feature MSE, so negative favors ours. The primary control metric averages all ten query predictions. Endpoint metrics and mean-five errors are secondary. Intervals jointly resample the three training seeds and recording-session clusters (10,000 draws); they are exploratory and unadjusted for multiple comparisons. † marks intervals including zero. Green bold point gains alone do not establish a difference.

Method	h	Original h5	Trained h10	Gain (%)	Paired 95% interval
Framewise	5	0.10972	0.10962	+0.09[†]	[-0.00028, +0.00007]
Framewise	10	0.15346	0.15261	+0.56	[-0.00135, -0.00039]
Constant dynamics	5	0.10968	0.10961	+0.07[†]	[-0.00025, +0.00009]
Constant dynamics	10	0.15338	0.15258	+0.52	[-0.00130, -0.00034]
ShiftWM (ours)	5	0.10938	0.10951	-0.12 [†]	[-0.00022, +0.00059]
ShiftWM (ours)	10	0.15385	0.15222	+1.06	[-0.00272, -0.00080]
Action-free	5	0.11011	0.11002	+0.08[†]	[-0.00022, +0.00005]
Action-free	10	0.15403	0.15322	+0.52	[-0.00114, -0.00046]

Table 4: **Matched training-horizon control: mean errors.** Original h5-trained and new h10-trained checkpoints use the exact same h10-eligible episode/session identities and query starts. The h5 rows use the five-step prefix of those windows; h10 uses all ten. These rows average all query steps. Gains are $100(\text{original} - \text{h10})/\text{original}$; paired intervals are h10 minus original. Both selection objectives retain window weighting but use their respective training horizons. [†] marks intervals including zero; green bold identifies only positive point gains. All four methods are shown, with no independent-test claim.

Method	h	Original h5	Trained h10	Gain (%)	Paired 95% interval
Framewise	5	0.15671	0.15630	+0.26	[-0.00081, -0.00002]
Framewise	10	0.21690	0.21429	+1.21	[-0.00383, -0.00148]
Constant dynamics	5	0.15664	0.15627	+0.23[†]	[-0.00076, +0.00002]
Constant dynamics	10	0.21675	0.21423	+1.16	[-0.00375, -0.00138]
ShiftWM (ours)	5	0.15623	0.15600	+0.15[†]	[-0.00090, +0.00061]
ShiftWM (ours)	10	0.21951	0.21342	+2.77	[-0.00936, -0.00309]
Action-free	5	0.15749	0.15709	+0.25	[-0.00067, -0.00010]
Action-free	10	0.21743	0.21496	+1.14	[-0.00323, -0.00168]

Table 5: **Matched training-horizon control: endpoint errors.** Original h5-trained and new h10-trained checkpoints use the exact same h10-eligible episode/session identities and query starts. The h5 rows use the five-step prefix of those windows; h10 uses all ten. These rows score the final query step. Gains are $100(\text{original} - \text{h10})/\text{original}$; paired intervals are h10 minus original. Both selection objectives retain window weighting but use their respective training horizons. [†] marks intervals including zero; green bold identifies only positive point gains. All four methods are shown, with no independent-test claim.

Method	Best epochs	Total (M)	Trained (M)	Seconds/run
Framewise	1/1/1	3.763	3.469	249 [242, 336]
Constant dynamics	1/1/1	3.763	3.714	262 [256, 368]
ShiftWM (ours)	1/1/1	3.763	3.714	264 [258, 368]
Action-free	1/1/1	3.763	3.312	234 [229, 321]

Table 6: **Horizon-ten checkpoints and computation.** Best epochs follow seeds 0/1/2, selected by window-weighted FP32 validation MSE over all ten recursive predictions after complete 30-epoch training. Total parameters include inactive frozen modules; trained parameters exclude them. Seconds/run is the median [minimum, maximum] successful runner invocation, including training, five evaluation passes and CPU reload checks, excluding prior interrupted work. Hardware partitions differ, so invocation times are operational records, not a matched speed comparison. All 12 selected packages pass registered and independently repeated weights-only CPU reload checks.